Technical Specification: Text Chunking Strategy

1. Algorithmic Overview

Large-scale documents containing institutional policies, manual configurations, or standard operating procedures cannot be effectively modeled as single monolithic text objects. High token density degrades semantic clarity, and exceeds the context windows of downstream LLM systems. This module breaks files down into dense, topical segments while preserving surrounding context.

2. Core Function:

chunk_text(text, max_chars=500)

The system implements a deterministic, token-aware character splitting algorithm designed around word boundaries. Instead of cutting strings off arbitrarily mid-word, the function processes the raw input string as an ordered array of tokens. It is a module part of the docs_import.py file.

3. Operational Mechanics

• Initialization: Tokenizes text via space-separation parameters while preserving system encodings.

• Accumulation Loop: Iteratively appends tokens to an active memory buffer while tracking real-time layout length.

• Constraint Boundary Evaluation: When the tracking length reaches or crosses the 500-character max_chars ceiling, the buffer is finalized as a string block.

• Remainder Resolution: Flushes residual token sequences into a trailing block, ensuring zero information loss.

4. Design Parameters

• max_chars (500): Optimally balanced to retain context without splitting key legal and operational concepts across multiple vectors.

• Word-Boundary Safety: Eliminates word truncation, ensuring high fidelity during vectorization transformations.